E-FILING 3.0 & 7-VECTOR JUDICIAL MATRIX

🏛️ AI-Powered Judicial Case Prioritization System

Comprehensive Technical Architecture & Stack Explanation Guide

Prepared for Project Reviewers, Technical Evaluators, & Mentors

📌 1. Executive Overview

The **AI-Powered Court System** is an enterprise-grade judicial management platform designed for District & Sessions Courts and High Courts. It solves chronic court backlog and hearing delays by integrating a **7-Vector Judicial Priority Matrix** powered by AI. The system automates lawsuit registration (under TN eFiling 3.0 standards), predicts case delay timelines, auto-schedules court hearing rosters, manages legal documents, and provides real-time judicial analytics.

> **Core Project Mission:** To eliminate manual, arbitrary court scheduling and ensure priority justice for vulnerable litigants (senior citizens, undertrials, and victims of heinous crimes) while maintaining 100% system uptime.

🛠️ 2. Technology Stack Breakdown

Layer	Technology	Role & Primary Purpose	Why It Was Chosen
Frontend UI	**React 18**	Single Page Application (SPA) framework for dynamic court dashboards.	Enables reusable, modular components and fast Virtual DOM re-rendering for high-interactivity legal forms.
Build Tool	**Vite 5**	Frontend build tool and dev server.	Provides instantaneous Hot Module Replacement (HMR) and optimized production JS bundling.
Styling	**Tailwind CSS**	Utility-first CSS styling framework.	Allows building a sleek, responsive dark/light mode interface with glassmorphism and judicial color palettes.
Data Visualization	**Chart.js & React-ChartJS-2**	Interactive chart rendering library.	Displays real-time court disposal metrics, pendency trends, and judicial performance graphs.
HTTP Client	**Axios**	Asynchronous REST API HTTP client.	Handles API requests, JSON serialization, bearer token authentication headers, and document upload progress.
Backend Engine	**Node.js & Express.js**	Server runtime and web API framework.	Provides a lightweight, event-driven, non-blocking I/O API gateway capable of handling high request volume.
Database	**PostgreSQL**	Relational Database Management System.	Guarantees strict ACID compliance, complex foreign key constraints, and relational queries across 9 core tables.
DB Connection	**`pg` (node-postgres)**	PostgreSQL driver with connection pooling.	Recycles database socket connections to prevent connection starvation under concurrent user load.
Security & Auth	**JWT & Bcrypt.js**	Token authentication & password encryption.	Uses one-way salted password hashing and stateless JWT bearer tokens for role-based security.
AI Microservice	**Groq LLM + Local Engine**	Dual-engine AI case priority calculator.	Delivers sub-second natural language legal case analysis with an automatic offline local fallback.

⚖️ 3. The 7-Vector Judicial Priority Matrix

When a case is registered, the AI engine evaluates lawsuit details against 7 distinct legal vectors to produce a **Priority Score (0 to 100)**, a **Priority Category (High / Medium / Low)**, and an estimated **Hearing Delay Timeline**.

Vector 1 (V1): Litigant Vulnerability Assessment

Evaluates if the petitioner is a Senior Citizen (Age ≥ 60 or 75), Differently-Abled (PwD), Terminally Ill, Minor, or Indigent. Gives immediate priority to ensure speedy justice for vulnerable individuals.

Vector 2 (V2): Personal Liberty & Undertrial Detention

Checks if the accused is in Judicial Custody and calculates detention duration. Automatically triggers Section 479 BNSS (statutory half-sentence rule) to prevent unlawful prolonged detention.

Vector 3 (V3): Gravity of Offense & Statutory Impact

Analyzes statutory offenses (e.g., POCSO Act 2012, Domestic Violence, IPC 302 Murder / 376 Rape, SC/ST Prevention of Atrocities Act). Prioritizes severe criminal matters impacting public safety.

Vector 4 (V4): Irreparable Harm & Emergency Relief

Detects urgent legal prayers including Eviction/Demolition Stay Orders, Injunctions, Habeas Corpus, or Urgent Bail to prevent irreversible damage before trial completion.

Vector 5 (V5): Case Pendency & Backlog Age

Evaluates how long the lawsuit has been pending (>1 year, >5 years, or >10 years) to target legacy backlog reduction.

Vector 6 (V6): Trial Stage Efficiency

Measures trial progress. Cases at the *"Final Arguments"* or *"Judgment"* stage receive priority points to encourage swift disposal.

Vector 7 (V7): Subject Matter & Commercial Valuation

Assesses financial stakes in commercial land, contract, and shipping disputes so economic suits move efficiently without stalling trade.

🔄 4. System Architecture & Data Flow

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+-------------------------------------------------------------------------------+ | SYSTEM ARCHITECTURE | +-------------------------------------------------------------------------------+ | [React 18 Frontend] ---> [Node.js Express API] ---> [PostgreSQL Database] | | (Client Port 5173) (Server Port 5000) (Users, Cases, Roster) | | | | | +---> [AI Microservice Engine (Port 5001)] | | | | | +------------+------------+ | | | | | | v v | | [Groq Cloud LLM API] [Local Heuristic Fallback] | | (7-Vector Prompt) (Offline Local Engine) | +-------------------------------------------------------+
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👥 5. Role-Based Access Control (RBAC)

User Role	Case Registration	Docket Editing	Assigned Bench Roster	Analytics Reports
Administrator	✅ Full Access	✅ Full Edit / Delete	✅ All Court Benches	✅ Full System Access
Hon'ble Judge	👁️ Read-Only	👁️ Read-Only	✅ Assigned Courtroom	✅ Bench Reports
Court Clerk	✅ Full Access	✅ Edit Case Details	✅ Roster Scheduling	✅ Filing Analytics